The Time Machine Credit Model

How fast does a loan approval model go out of date, and what warns you first?

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Contents

1 Abstract 3

2 Introduction and problem statement 3

3 Data 4

4 Exploratory data analysis 6

5 Methodology 9

6 Results: the scorecard 12

7 Results: the frozen model through time 12

8 Results: early-warning signals 14

9 Results: comparing fixes 15

10 Discussion 16

11 Limitations 17

12 Conclusion and recommendations 17

13 References 18

A Glossary 19

B Reproducibility 19

C Full results and unresolved placeholders 20

List of Figures

1 Synthetic benchmark bad rate by origination vintage. 6
2 Synthetic benchmark origination volume by year. 7
3 Missing-value profile across candidate predictors. 8
4 Synthetic benchmark bad rate by FICO score band. 8
5 Synthetic benchmark bad rate by loan purpose. 9
6 Frozen scorecard Gini by origination year. 13
7 Frozen scorecard calibration gap by origination year. 13
8 Characteristic Stability Index by selected predictor and year. 14
9 Modeled profit per applicant under the three remediation policies. 16

List of Tables

1 Observed synthetic benchmark cohort by origination year 4
2 Candidate variable dictionary 5
3 Benchmark model card: selected predictors and Information Value 12
4 Scorecard detail still required for a fully specified model card 12
5 Frozen benchmark scorecard performance by vintage 12
6 Benchmark PSI and formal surveillance alerts 14
7 Benchmark remediation strategies, 2016–2018 15
8 Economic sensitivity: average profit per applicant by scenario 16

1 Abstract

This report asks how quickly a credit approval scorecard loses usefulness, which monitoring signal warns first, and whether a cut-off reset or a model rebuild is the better response. The supplied project package did not include the raw Lending Club loan file; its reproducible fallback instead generated a 60,000-record synthetic benchmark for 2012–2018. A weight-of-evidence logistic scorecard was fitted on 2012–2014, checked on 2015, and frozen for later vintages. In that benchmark, Gini was 0.329 in 2015 and ranged from 0.317 to 0.344 through 2018, with no year meeting the specified 0.050 Gini-decay rule. Calibration moved more sharply: the predicted-to-observed default gap reached 20.9% in 2017 and 40.0% in 2018. Score PSI remained low, rising from 0.002 to 0.023, so the 2018 alert came from calibration rather than score-distribution PSI. In the base economic scenario, annual cut-off reset averaged \$653.06 profit per applicant versus \$650.45 for doing nothing and \$640.09 for rebuilding; the ranking changed under stressed assumptions. These are benchmark results, not empirical Lending Club findings. They illustrate why discrimination, calibration, population drift, and outcome maturity must be monitored together; real-data conclusions require rerunning the pipeline on the Kaggle file.

Key takeaway. On the supplied synthetic run, calibration diverged while rank-order discrimination remained broadly stable; none of these figures should be interpreted as measured Lending Club performance.

2 Introduction and problem statement

This section frames the monitoring problem, states the questions, and explains why model performance can change after deployment. The analysis is a controlled demonstration using the project package's synthetic fallback, not a real-loan back-test.

Credit approval models are developed on historical applicants and later applied to new applicant populations. Economic conditions, customer mix, product design, and underwriting policy can change. A model that once separated relatively safer borrowers from riskier borrowers may continue to rank applicants sensibly while its predicted default probabilities become too high or too low. A decline in model quality over time is commonly called **model decay**; it can affect both risk estimates and business decisions.

The project asks three questions:

1. How quickly do predictive ranking, calibration, and the modeled profit measure change over time?
2. Which monitored indicator gives the earliest useful warning: score or feature drift, calibration, or discrimination?
3. Which tested response—keeping the existing policy, resetting the approval cut-off, or rebuilding the scorecard—performs best under stated assumptions?

These questions matter to model development, independent validation, ongoing monitoring, and credit strategy. Development establishes a defensible baseline; validation tests whether the model generalizes to a later period; monitoring identifies changes after implementation; and strategy determines what action is proportionate to the evidence. Since the supplied results are synthetic, they can demonstrate this workflow but cannot establish the behavior of an actual bank or Lending Club portfolio.

2.1 Evidence boundary

The source scripts contain a fallback generator that creates 60,000 simulated records when the raw CSV is absent. The packaged tables and figures correspond to that fallback: the yearly counts sum to 60,000, and the generator explicitly imposes changing feature distributions and reduced observed

bad probabilities for the later years. The Kaggle CSV was not included in the uploaded bundle. Therefore, all numerical results below are labeled as synthetic benchmark results. No statement about actual 2007–2018 accepted loans is inferred from them.

Key takeaway. This is a reproducible demonstration of an out-of-time monitoring design; it is not an empirical Lending Club study until the raw source data are supplied and the pipeline is rerun.

3 Data

This section describes the intended source and the data actually used by the supplied run. The distinction is essential because the benchmark generator does not perform the real-loan filters described in the ingestion script.

3.1 Intended source and observed run

The project is designed for the accepted-loan file from the Lending Club dataset curated by Nathan George on Kaggle, covering loans issued from 2007 through 2018Q4 and distributed under CC0. The project README identifies the raw file as `accepted_2007_to_2018Q4.csv.gz`. That file was not present in the supplied package. The pipeline therefore used its built-in synthetic cohort: 60,000 records with issue years 2012–2018. The displayed counts, bad rates, charts, model metrics, and economic outputs all describe this generated cohort only.

In the intended raw-data workflow, the scripts retain 36-month loans with observed outcomes `Fully Paid` or `Charged Off`; define `bad = 1` for `Charged Off` and `bad = 0` for `Fully Paid`; and discard other statuses. The synthetic fallback directly generates a binary target and does not apply those loan-term/status filters. Thus the raw-data cohort size, filters actually applied to Lending Club, and target prevalence in the intended dataset are not established by the supplied benchmark outputs.

Table 1: Observed synthetic benchmark cohort by origination year

Issue year	Records	Bad outcomes	Bad rate	Mean FICO / DTI
2012	4,965	483	9.7%	696.0 / 18.0
2013	7,166	711	9.9%	694.7 / 18.3
2014	10,765	1,080	10.0%	695.0 / 18.9
2015	13,425	1,379	10.3%	695.0 / 19.1
2016	10,565	1,110	10.5%	695.0 / 19.7
2017	7,053	619	8.8%	694.8 / 20.0
2018	6,061	470	7.8%	694.7 / 20.4

FICO is the generated score midpoint; DTI is the generated debt-to-income ratio. Mean FICO and DTI are shown as FICO / DTI.

Key takeaway. Table 1 totals 60,000 synthetic records; the lower recorded bad rates in 2017–2018 are consistent with the generator's explicit seasoning adjustment and must not be treated as observed loan maturity behavior.

3.2 Candidate variables and leakage controls

The raw-data script retained variables intended to be available at application time and derived a FICO midpoint, numeric employment length, and credit-history length. It excluded Lending Club's proprietary interest rate and grades, as well as post-origination payment, recovery, and collection information. These exclusions reduce target leakage: leakage occurs when a predictor contains information unavailable at the decision time or information created after the outcome process began. The candidate dictionary below reflects the feature list constructed by the cleaning script; categorical/numeric distinctions describe intended use and do not imply each field was present in the synthetic fallback in the same form.

Table 2: Candidate variable dictionary

Variable	Description and reason for retention	Type
`loan_amnt`	Requested principal; represents requested exposure.	Numeric
`emp_len`	Employment length in years; employment stability proxy.	Numeric
`home_ownership`	Housing tenure category.	Categorical
`annual_inc`	Reported annual income; capacity proxy.	Numeric
`verification_status`	Whether income was verified.	Categorical
`purpose`	Applicant-stated purpose for borrowing.	Categorical
`addr_state`	State of residence; geographic portfolio descriptor.	Categorical
`dti`	Debt-to-income ratio; existing debt burden relative to income.	Numeric
`delinq_2yrs`	Reported delinquencies over the previous two years.	Numeric
`inq_last_6mths`	Recent credit inquiries; recent credit-seeking activity.	Numeric
`open_acc`	Number of open credit accounts.	Numeric
`pub_rec`	Number of public-record indicators.	Numeric
`revol_bal`	Revolving credit balance.	Numeric
`revol_util`	Revolving balance relative to available revolving credit.	Numeric
`total_acc`	Total reported credit accounts.	Numeric
`mort_acc`	Number of mortgage accounts.	Numeric
`pub_rec_bankruptcies`	Reported public-record bankruptcies.	Numeric
`fico`	Midpoint of the reported low and high FICO range.	Numeric
`credit_hist_yrs`	Issue year less earliest reported credit-line year.	Numeric

Key takeaway. Table 2 lists 19 candidate application-time features; only four passed the run's IV filter and entered the benchmark scorecard.

3.3 Chronological design

The design used 2012–2014 for model building, 2015 for validation and cut-off selection, and 2016–2018 for forward testing. A random split would mix years between development and test data, allowing the test set to share time-specific conditions with the training sample. A chronological split more closely represents deployment, where a model is estimated using the past and then applied to future cases.

Key takeaway. The time split is appropriate for a forward-looking model test, but its validity for the real Lending Club portfolio remains untested until the raw data are run through the supplied ingestion path.

4 Exploratory data analysis

This section reviews cohort volume, missingness, borrower risk patterns, and the limits of outcome maturity. The five provided exploratory charts are embedded unchanged and interpreted below.

4.1 Volume and bad rate over time

Figure 1 shows the benchmark bad rate by issue year, while Figure 2 shows that the generated sample is largest in 2015 and becomes smaller in the final two years. The 2017–2018 bad-rate decline is not independent evidence of improving credit quality: the data-generation code explicitly multiplies the simulated default probability by 0.85 in 2017 and 0.65 in 2018. It is a modeled seasoning adjustment, not a measurement of actual repayment maturity. The supplied Figure 1 subtitle also states that defaults stabilize around 15–18%, which conflicts with Table 1 and its plotted values of 9.7–10.5% for 2012–2016. The chart is preserved unchanged; this report follows the numeric table and plotted series rather than that subtitle.

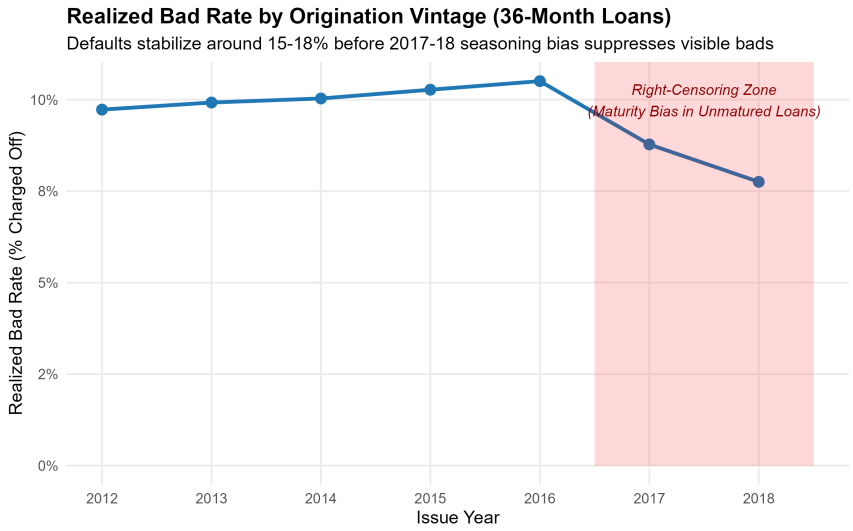


Figure 1: Synthetic benchmark bad rate by origination vintage.

Key takeaway. The plotted downturn after 2016 is partly built into the synthetic target-generation rule; it cannot establish a real-world late-vintage default trend.

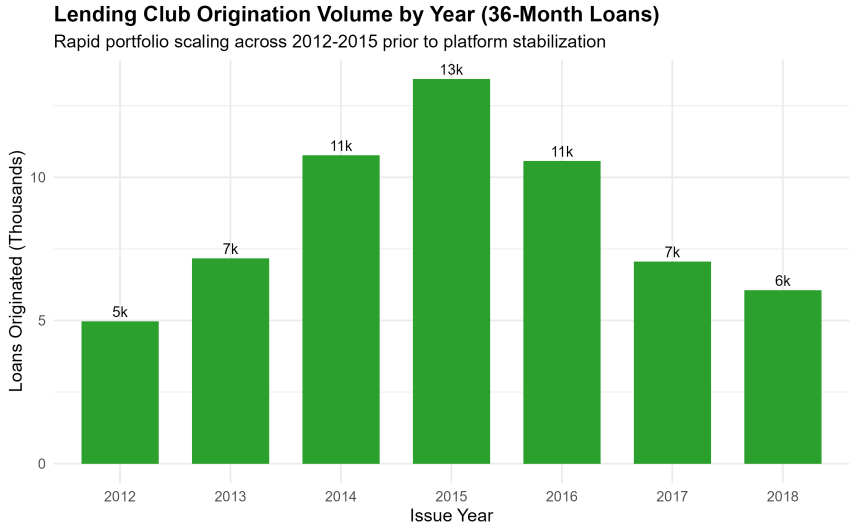


Figure 2: Synthetic benchmark origination volume by year.

Key takeaway. Cohort sizes vary materially, so year-to-year rates have different sample bases; the figure describes the generated sample rather than the Lending Club portfolio.

4.2 Missingness and risk segmentation

Figure 3 presents the missing-value profile for candidate predictors. Missingness is concentrated in a small subset of fields in the supplied visualization, and the modeling workflow relies on the binning process to handle missing categories rather than silently replacing them with a typical value. Exact feature-level missing percentages are not available as a printed table in the delivered results and are not transcribed from the plot.

Figure 4 groups the benchmark's realized bad rate by FICO band. The chart shows a broadly declining pattern as FICO rises, matching the simulated target equation's negative FICO-risk relationship. This pattern is therefore a property of the generator as well as of the fit; it is not independent evidence of the Lending Club data's risk gradient. Figure 5 compares outcome rates across loan-purpose categories. The visible differences motivate segment monitoring, but the supplied report does not provide the underlying purpose-level counts and rates in a numerical table.

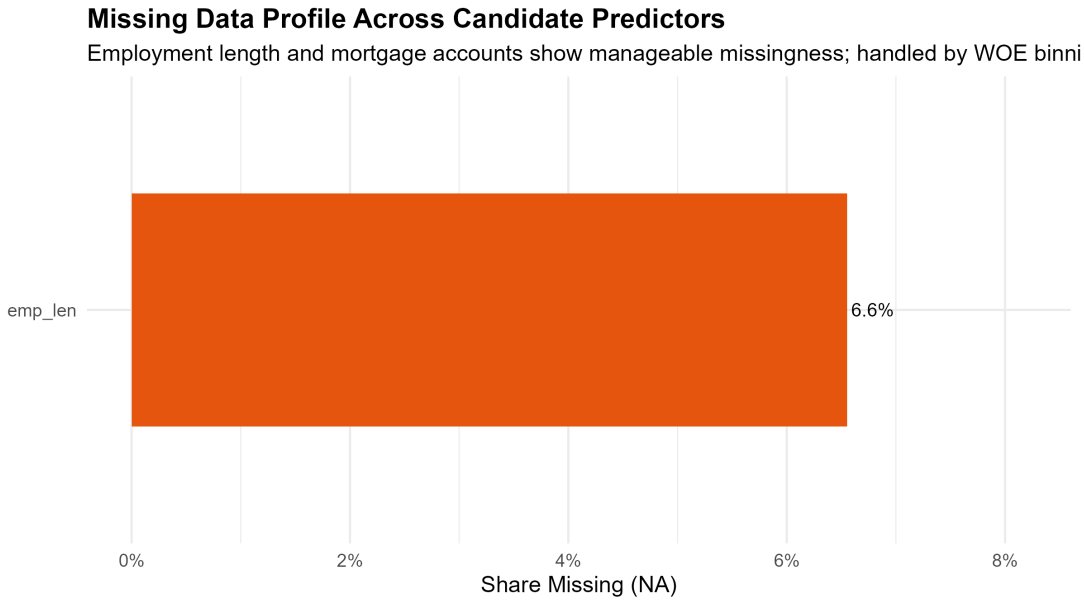


Figure 3: Missing-value profile across candidate predictors.

Key takeaway. Missingness is uneven across candidate variables; preserve missingness as a bin or explicitly document imputation rather than assume all inputs are complete.

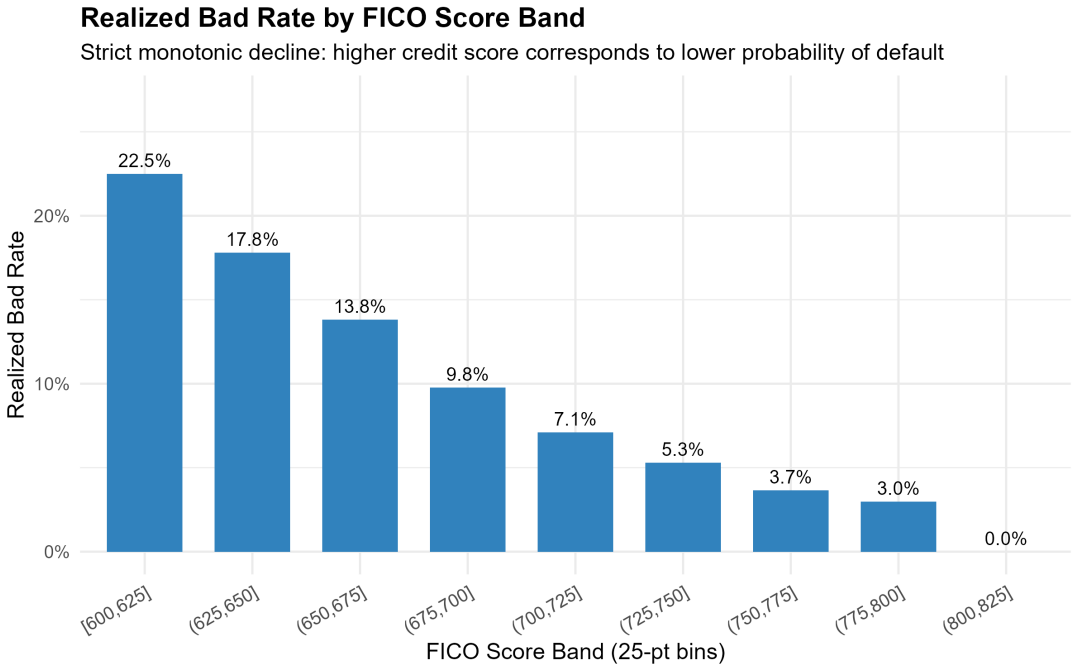


Figure 4: Synthetic benchmark bad rate by FICO score band.

Key takeaway. The chart shows the expected inverse FICO-risk pattern, but its presence is partly guaranteed by the way the synthetic bad outcome was generated.

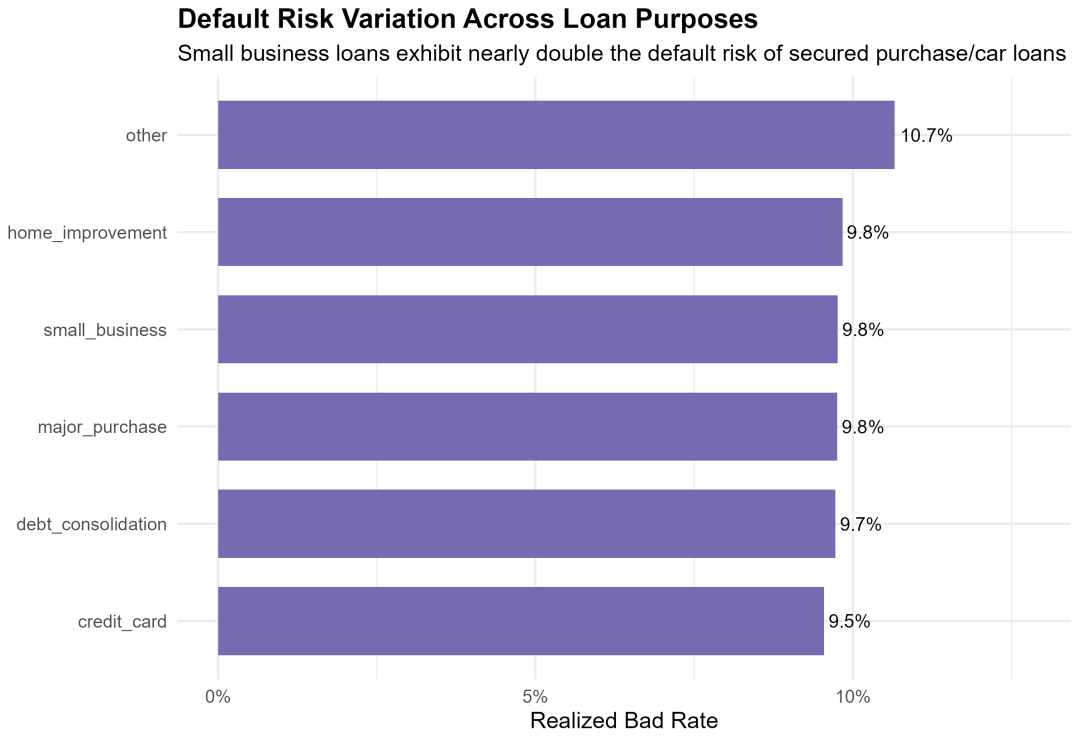


Figure 5: Synthetic benchmark bad rate by loan purpose.

Key takeaway. Outcome rates differ across purposes in the chart, supporting segment-level monitoring; exact subgroup rates require the underlying results table.

4.3 Right-censoring and seasoning

Right-censoring occurs when a loan has not had enough time to experience the event being measured, so a non-default observation may later become a default. The intended dataset ends at 2018Q4 and the outcome definition requires completed 36-month loans. However, the included fallback is a synthetic generator, not a survival dataset: it deliberately reduces recorded bad probability for 2017 and 2018 to mimic seasoning. Its target rates therefore cannot quantify actual right-censoring or estimate lifetime default. A real-data rerun should use mature cohorts or a time-to-event method and disclose observation windows.

Key takeaway. Late-vintage outcomes are not mature evidence in this benchmark; the apparent improvement is mechanically imposed and should not guide credit policy.

5 Methodology

This section explains the model construction, score scaling, performance metrics, drift checks, and the core code. The supplied run used a weight-of-evidence logistic scorecard, then held the 2012–2014 fit fixed for the forward performance analysis.

5.1 WOE binning and information value

Weight of Evidence (WOE) replaces a raw predictor with a value representing the relative concentration of good and bad outcomes in a bin. For bin i, under the common good-to-bad convention,

$$\mathrm{WOE}_i = \ln\left(\frac{G_i/G}{B_i/B}\right), \qquad \mathrm{IV} = \sum_i \left(\frac{G_i}{G} - \frac{B_i}{B}\right)\mathrm{WOE}_i,$$

where G_i and B_i are the good and bad counts in bin i, and G and B are the total good and bad counts. Larger absolute WOE indicates stronger separation in that bin; IV aggregates separation across bins. The implementation used tree-based binning and retained variables with IV at least 0.02. The package's internal WOE sign convention should be checked against its version before interpreting coefficient direction.

The delivered report does not include the per-bin good/bad counts, cut boundaries, or WOE values. A numeric worked example would therefore require inventing inputs, so none is fabricated. In a completed empirical run, those quantities should be inserted from the fitted binning object as `[INSERT BIN COUNTS, BOUNDARIES, AND WOE EXAMPLE]`.

5.2 Logistic regression and score scaling

The model estimates the probability of a bad outcome from the WOE-coded features using logistic regression:

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \sum_j \beta_j \,\mathrm{WOE}_j, \qquad p = \frac{1}{1+e^{-\eta}}.$$

Here p is predicted probability of default and η is the linear predictor. Holding other terms fixed, a coefficient indicates how a one-unit change in that feature's WOE changes the log-odds. Its sign must be read together with the package's WOE convention. The exact fitted coefficients and standard errors were not present in the printed result tables; they remain `[INSERT MODEL COEFFICIENTS AND STANDARD ERRORS]`.

The scorecard uses 600 points at bad-to-good odds 1/19 and a points-to-double-odds (PDO) value of 50. PDO is the score increase associated with halving the odds of default. The scale constants are

$$\mathrm{Factor} = \frac{\mathrm{PDO}}{\ln 2} = 72.135, \qquad \mathrm{Offset} = 600 + \mathrm{Factor}\ln(1/19) = 387.604,$$

with score expressed as $\mathrm{Offset} - \mathrm{Factor}\ln(\text{bad odds})$. A higher score therefore corresponds to lower bad odds under this convention. A cut-off of approximately 533 points was selected on 2015 validation to target a 70% approval share; the observed validation approval rate was 70.1%.

5.3 Performance and economics

Discrimination is the ability to rank bad cases above good cases by risk, regardless of a chosen cut-off. The Area Under the ROC Curve (AUC) summarizes ranking over all cut-offs, and Gini is $2\,\mathrm{AUC} - 1$. The Kolmogorov–Smirnov (KS) statistic is the maximum separation between cumulative score distributions for goods and bads:

$$\mathrm{KS} = \sup_s \left| F_{\mathrm{good}}(s) - F_{\mathrm{bad}}(s) \right|.$$

The 2015 KS result is not in the delivered result table and is marked `[INSERT 2015 KS]`. Approval rate is $N^{-1}\sum_i I(A_i = 1)$, where A_i indicates approval; bad rate among approved is $\sum_i I(A_i = 1)B_i / \sum_i I(A_i = 1)$, where B_i is the bad indicator.

Calibration measures agreement between average predicted probability and observed default frequency. The run defines

$$\text{Calibration Gap} = \frac{\text{Mean PD}}{\text{Observed Bad Rate}} - 1.$$

A positive gap means average predicted probability exceeds the observed bad rate; a negative gap means it is lower. The sign matters, and the estimate is unreliable when outcomes are immature or the target is synthetically altered.

Approval rate is the share of applicants with score at or above the selected cut-off. **Bad rate among approved** is the observed fraction of bad outcomes in that approved subset. The stylized profit calculation assumes a good funded loan earns a margin of 10% of loan amount and a bad loan incurs loss given default (LGD) of 60% of loan amount:

$$\text{Profit per applicant} = \frac{1}{N}\sum_{i=1}^{N} I(\text{approved}_i)\,\text{loan amount}_i \times \begin{cases} 0.10, & \text{if good,} \\ -0.60, & \text{if bad.} \end{cases}$$

This is a scenario metric, not booked profit, and omits funding costs, expenses, recoveries beyond the LGD assumption, and capital costs. The outputs are in dollars per applicant; conversion to a percentage of average loan amount is unavailable because that average was not included in the printed outputs: `[INSERT PROFIT AS % OF AVERAGE LOAN AMOUNT]`.

5.4 Population and characteristic stability

Population Stability Index (PSI) summarizes the change in a binned score distribution between a reference population and a later population:

$$\text{PSI} = \sum_i (A_i - E_i)\ln\left(\frac{A_i}{E_i}\right),$$

where E_i and A_i are the reference and later proportions in bin i. Characteristic Stability Index (CSI) applies the same idea to a single predictor's bins. PSI/CSI detect changes in input distributions; by themselves they do not show that risk relationships or outcomes have worsened.

The project uses conventional guide bands: PSI below 0.10 is broadly stable, 0.10–0.25 merits investigation, and above 0.25 indicates material population shift. The formal alert rule is PSI > 0.25 or absolute calibration gap > 0.25; a model is labeled decayed only if Gini falls by more than 0.05 versus 2015. These are project thresholds, not universal regulatory cut-offs.

5.5 Core R workflow

The following short listing shows the main analytical steps; the full scripts remain in the supplied project folder.

Listing 1: Core scorecard and monitoring functions from the supplied R pipeline.

```
fit_scorecard <- function(train, iv_min = 0.02) {
  bins <- woebin(train, y = "bad", method = "tree")
  iv <- sapply(bins, function(x) x$total_iv[1])
  kept <- names(iv[iv >= iv_min])
  woe <- woebin_ply(train[, c(kept, "bad")], bins[kept])
  model <- glm(bad ~ ., data = woe, family = binomial())
  card <- scorecard(bins[kept], model,
                    points0 = 600, odds0 = 1/19, pdo = 50)
  list(bins = bins[kept], model = model, card = card)
}

calib_gap <- mean(predicted_pd) / mean(observed_bad) - 1
gini <- 2 * as.numeric(pROC::auc(pROC::roc(bad, pd))) - 1
alert <- (psi > 0.25) | (abs(calib_gap) > 0.25)
decayed <- (gini_2015 - gini_current) > 0.05
```

Key takeaway. The method is interpretable and reproducible in structure, but the unavailable bin details, coefficient output, and run environment limit full independent reproduction of the fitted model from this report alone.

6 Results: the scorecard

This section summarizes the variables retained by the IV filter and the validation benchmark. The model card is available in the generated report; detailed point contributions and full coefficients are not.

Table 3: Benchmark model card: selected predictors and Information Value

Variable	IV	Reported strength
FICO midpoint (`fico`)	0.2070	Medium
Debt-to-income (`dti`)	0.1227	Medium
Revolving utilization (`revol_util`)	0.0741	Weak
Recent inquiries (`inq_last_6mths`)	0.0232	Weak

IV strength labels follow the supplied report's bands. The build period was 2012–2014.
Key takeaway. Four predictors passed IV ≥ 0.02; FICO had the largest IV (0.2070) in this synthetic run.

The 2015 validation cut-off was approximately 533 points, and the printed performance table shows a 70.1% approval rate. The supplied deck reports a 70% target; the exact 2015 KS statistic and model coefficients remain missing. Figure 6 visualizes Gini through time, including the validation observation, and helps distinguish rank stability from calibration drift.

Table 4: Scorecard detail still required for a fully specified model card

Requested detail	Status
2015 validation KS	`[INSERT 2015 KS]`
Score cut-off and validation rate	533 points; 70.1% reported approval rate
Bin boundaries, counts, WOE and points	`[INSERT BIN/POINTS TABLE FOR TOP VARIABLES]`
Model coefficients and standard errors	`[INSERT MODEL COEFFICIENTS]`

Key takeaway. Table 4 separates supplied benchmark results from model-card outputs that were not included and must not be guessed.

7 Results: the frozen model through time

This section reports the benchmark's frozen-scorecard performance from validation through the three forward years. Gini remained within a narrow observed range, while the calibration gap increased markedly in 2017–2018.

Table 5: Frozen benchmark scorecard performance by vintage

Year	Loans	Bad rate	Mean PD	Calibration gap	Gini	Approval	Bad among approved	Profit / applicant
2015	13,425	10.3%	10.2%	−1.2%	0.329	70.1%	7.3%	$531.77
2016	10,565	10.5%	10.4%	−0.9%	0.317	68.3%	7.3%	$531.14
2017	7,053	8.8%	10.6%	20.9%	0.344	67.5%	5.6%	$680.49
2018	6,061	7.8%	10.9%	40.0%	0.333	65.8%	4.8%	$739.71

Profit is the modeled dollar amount per applicant under MARGIN = 0.10 and LGD = 0.60; it is not profit as a percentage of loan amount.
Key takeaway. Table 5 shows no 0.050 Gini drop, but mean predicted PD exceeds observed bad rate by 40.0% in 2018; late-year target rates are affected by the synthetic seasoning rule.

Figure 6 shows the Gini trajectory directly. It fell from 0.329 in 2015 to a low of 0.317 in 2016, then

rose to 0.344 in 2017 and ended at 0.333 in 2018. The 2018 value is 0.004 above the 2015 baseline; no year satisfies the specified decay rule. Figure 7 shows that the calibration gap crosses the +25% threshold in 2018. Together, the charts say that calibration, not observed ranking power, is the primary deterioration signal in this benchmark.

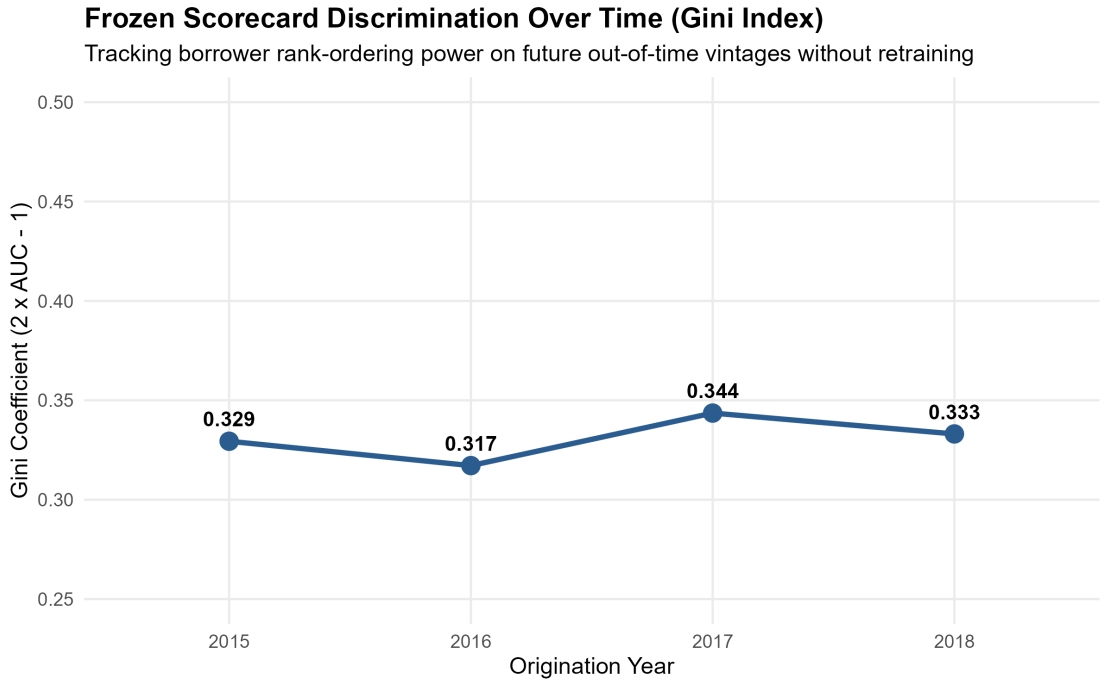


Figure 6: Frozen scorecard Gini by origination year.
Key takeaway. Gini ranged from 0.317 to 0.344 and finished 2018 above its 2015 level, so the run does not demonstrate Gini decay.

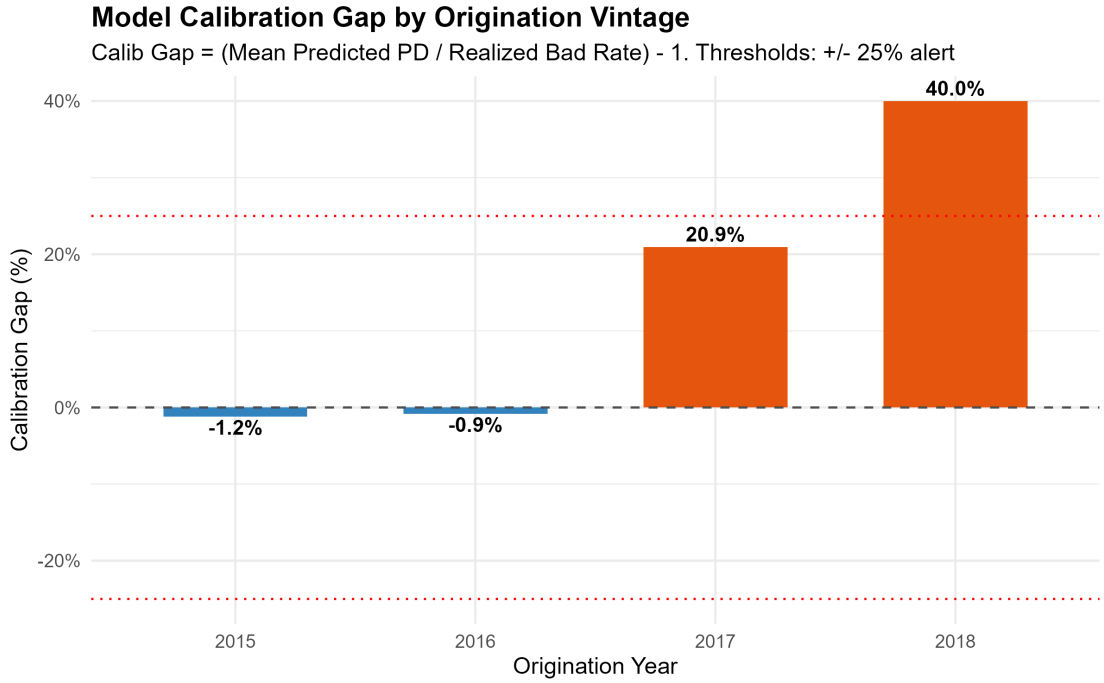


Figure 7: Frozen scorecard calibration gap by origination year.
Key takeaway. The calibration gap rises to 40.0% in 2018 and crosses the project's +25% alert limit; the benchmark's imposed seasoning effect is a key caveat.

Key takeaway. Ranking is resilient in the supplied run; calibration becomes the earlier and stronger warning, although synthetic outcome adjustment confounds interpretation of later vintages.

8 Results: early-warning signals

This section compares score-distribution stability, feature drift, and the alert rule. Score PSI stays below the 0.10 stable-band boundary in every reported year; calibration triggers the only formal alert, in 2018.

Table 6: Benchmark PSI and formal surveillance alerts

Year	Gini	Gini drop vs 2015	Calibration gap	PSI	Alert	Decayed
2015	0.329	0.000	−1.2%	0.0021	No	No
2016	0.317	0.012	−0.9%	0.0069	No	No
2017	0.344	−0.014	20.9%	0.0114	No	No
2018	0.333	−0.004	40.0%	0.0230	Yes	No

Alert = PSI > 0.25 or $|$calibration gap$| > 0.25$. Decayed = 2015 Gini less current Gini > 0.05.

Key takeaway. Table 6 shows the first formal alert in 2018, driven by calibration; PSI never approaches 0.10, and Gini never meets the decay rule.

Figure 8 shows the CSI heat map for the four selected predictors. The displayed CSI values help locate input mix changes, but PSI and CSI measure distribution shifts, not deterioration in predictive relationships. The full machine-readable CSI output table was not printed in the supplied report; its export is a remaining reproducibility item.

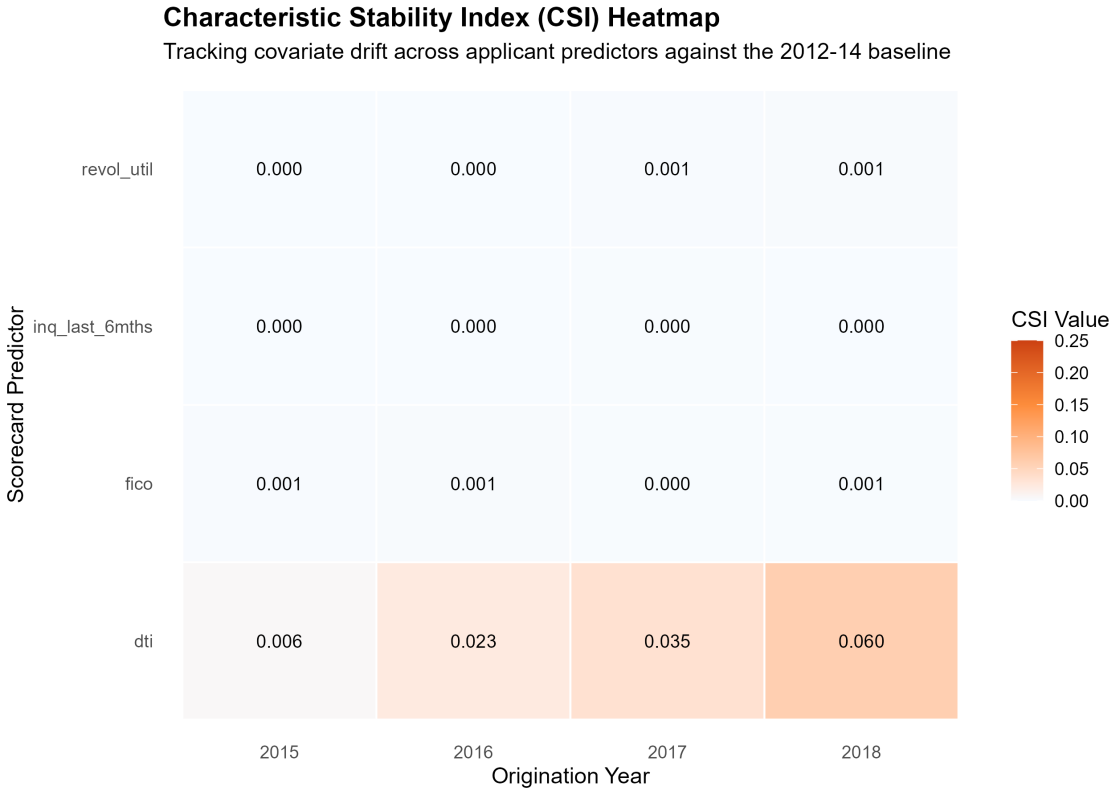


Figure 8: Characteristic Stability Index by selected predictor and year.

Key takeaway. CSI is a feature-level diagnostic that can show which inputs moved, but no feature-level alert threshold was applied in the formal alert rule supplied here.

Lead time. The alert fired in 2018 and no Gini deterioration event was observed through 2018; therefore, lead time to ranking deterioration is not measurable in the supplied run. The 2018 alert coincides with a calibration-gap breach rather than preceding a demonstrated Gini decline. PSI remained quiet while calibration drifted, illustrating that PSI alone can miss a change in the relationship between score and outcome.

Key takeaway. Monitor PSI/CSI and outcome-based measures together; this run contains no evidence that PSI warned before performance decay, because the PSI threshold never fired and Gini did not decay.

9 Results: comparing fixes

This section compares the three supplied policies and tests whether the profit ranking is robust to alternative margin and loss assumptions. No strategy dominates across every year and scenario.

Strategy A keeps the frozen scorecard and fixed 2015 cut-off. Strategy B retains the frozen model but recalculates the score threshold using the prior-year score distribution. Strategy C rebuilds the model on a rolling three-year window. Table 7 reports the supplied outputs; Figure 9 visualizes profit per applicant.

Table 7: Benchmark remediation strategies, 2016–2018

Year	Strategy	Approval	Bad among approved	Gini	Profit / applicant
2016	A: Do nothing	68.3%	7.3%	0.317	$531.14
2016	B: Reset cut-off	68.3%	7.3%	0.317	$531.14
2016	C: Rebuild	67.7%	7.4%	0.317	$521.27
2017	A: Do nothing	67.5%	5.6%	0.344	$680.49
2017	B: Reset cut-off	70.9%	6.0%	0.344	$691.39
2017	C: Rebuild	72.1%	6.1%	0.343	$691.28
2018	A: Do nothing	65.8%	4.8%	0.333	$739.71
2018	B: Reset cut-off	69.3%	5.3%	0.333	$736.65
2018	C: Rebuild	67.6%	5.5%	0.319	$707.73

Output values are benchmark dollars per applicant under MARGIN = 0.10 and LGD = 0.60.

Key takeaway. Table 7 shows the reset policy has the highest 2017 profit, while doing nothing leads in 2018; full rebuild is not the most profitable policy in any listed year.

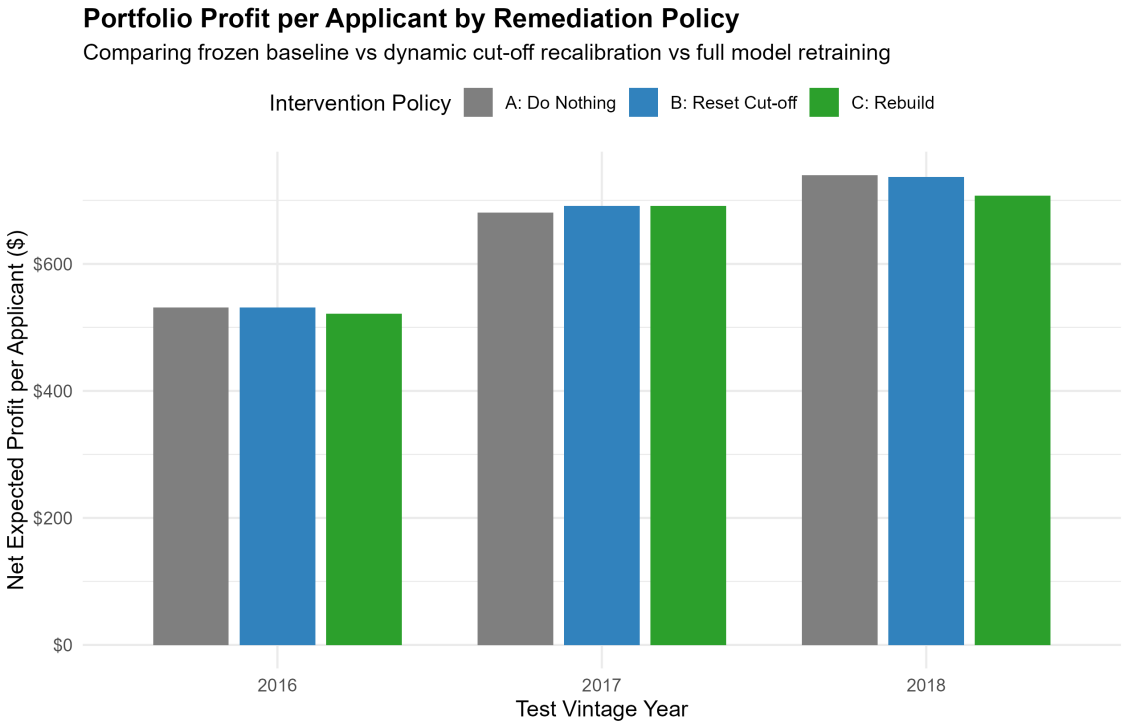


Figure 9: Modeled profit per applicant under the three remediation policies.
Key takeaway. Strategy B is competitive in this base case, but the differences are small relative to the assumptions and do not establish a universally optimal policy.

Table 8 reports average profits across 2016–2018 under three economic scenarios. Strategy B ranks first in the base and high-margin cases, but Strategy A ranks first under stressed loss assumptions. The economic ranking therefore does not hold across all tested assumptions.

Table 8: Economic sensitivity: average profit per applicant by scenario

Scenario	Margin / LGD	A: Do nothing	B: Reset cut-off	C: Rebuild
Base case	0.10 / 0.60	$650.45	$653.06	$640.09
Stressed loss	0.08 / 0.70	$374.33	$364.57	$352.80
High margin	0.12 / 0.50	$926.56	$941.55	$927.39

Key takeaway. Table 8 reverses the A-versus-B ranking under stressed loss assumptions, so the preferred action depends on validated economics rather than the base case alone.

Key takeaway. A cut-off reset is a reasonable low-cost option to test, not a proven winner; the sensitivity table argues against automatic resets without portfolio-specific economic validation.

10 Discussion

This section translates the benchmark findings into monitoring implications and explains the value and limits of a transparent scorecard. The practical lesson is to diagnose which performance dimension moved before selecting an intervention.

First, discrimination and calibration answer different questions. Gini measures relative ranking; calibration asks whether predicted probabilities match realized default rates. A stable Gini does not guarantee accurate probability estimates, capital forecasts, pricing, or loss planning. The benchmark makes this distinction visible: the 2018 Gini is 0.333, compared with 0.329 in 2015, while the calibration gap is 40.0%.

Second, population stability measures are not outcome measures. PSI remained at 0.0230 or below, even though the 2018 calibration gap breached its threshold. This can happen when score distributions remain similar but the relationship between applicant risk and observed outcomes shifts. CSI adds localization at feature level, but neither PSI nor CSI should be treated as proof of model decay without outcome evidence and a valid observation window.

Third, action should be proportionate. The tested Strategy B can adjust volume without re-estimating coefficients, but the sensitivity results show it does not always maximize the chosen profit metric. A model rebuild is more resource intensive and, in these benchmark values, does not dominate. For a bank, a threshold change still requires policy approval, impact analysis, monitoring of adverse outcomes, and compliance review; it is not merely a technical recalibration.

The scorecard is transparent: bins, WOE transformations, logistic coefficients, and score points can be inspected and explained. This supports governance, reason-code development, and independent validation. A more complex model could be tested as a challenger, but a black-box model should not replace a transparent champion without evidence on performance, stability, fairness, explainability, and operational controls.

Key takeaway. Monitoring policy should pair ranking, calibration, drift, and economics; no single chart or threshold can justify a remediation decision on its own.

11 Limitations

This section records the main limitations so the benchmark is not mistaken for a completed empirical credit study. Each limitation affects interpretation or external validity.

1. **Synthetic rather than raw data.** The uploaded package did not contain the Kaggle CSV. The cached benchmark is generated by code and contains 60,000 records, not observed Lending Club loans.
2. **Outcome maturity.** The generator imposes lower observed bad probabilities for 2017–2018. It does not model loan-level censoring, maturity dates, or survival times.
3. **No COVID-era vintages.** The project ends in 2018 and does not cover pandemic-era credit conditions.
4. **Accepted loans only in intended design.** The raw-data plan excludes rejected applicants; reject inference and selection effects are not addressed. The synthetic fallback does not represent an actual acceptance process.
5. **Assumed economics.** MARGIN and LGD are stylized parameters. The sensitivity analysis shows that strategy ordering changes with them.
6. **Proxy and policy changes.** Lending Club personal loans are not a universal bank portfolio, and its underwriting rules may have changed over time. Neither effect can be evaluated from this synthetic run.
7. **Missing model artifacts.** Exact coefficients, KS, bin boundaries, WOE values, points by bin, and package versions were not included in the printed results.

Key takeaway. The strongest limitation is the synthetic data source; until actual accepted-loan data are processed, conclusions are methodological illustrations only.

12 Conclusion and recommendations

This section answers the three project questions and gives a cautious monitoring recommendation. The numerical statements below are specific to the supplied synthetic benchmark and should be replaced after an empirical rerun.

The scorecard's Gini ranged from 0.317 to 0.344 between 2015 and 2018 and ended at 0.333, above the 2015 value of 0.329. Calibration changed more: its gap rose to 20.9% in 2017 and 40.0% in 2018, while PSI remained at or below 0.0230. The first formal alert fired in 2018 on calibration, and no year crossed the 0.050 Gini-decay threshold, so lead time to ranking deterioration is not measurable. Under base economics, Strategy B averaged \$653.06 per applicant versus \$650.45 for Strategy A and \$640.09 for Strategy C; under stressed loss economics, Strategy A ranked first. The figures are synthetic and do not quantify Lending Club's realized model decay.

Three-line recommendation

1. Use Strategy B as a monitored, low-cost cut-off option only after confirming current mature-cohort calibration, approval targets, and portfolio economics.
2. Investigate any |calibration gap| > 0.25 or PSI > 0.25; separate maturity/data-quality effects from genuine drift before changing policy.
3. Reserve a full rebuild for sustained evidence, such as Gini decline > 0.05 versus validation together with material drift or persistent calibration failure, followed by independent validation.

Monitoring plan. Review score PSI and feature CSI monthly or quarterly; track Gini/KS and calibration at each outcome refresh; report approval rate, approved bad rate, and realized portfolio economics alongside them. Use PSI 0.10 as a review/watch boundary and 0.25 as a material-drift trigger, calibration-gap magnitude 0.25 as an outcome alert, and Gini decline 0.05 as the specified decay threshold. Apply these thresholds only with adequate sample size and mature outcomes, and document escalation, ownership, and sign-off.

Future work should include reject inference with appropriate sensitivity analysis, a survival/lifetime default model for incomplete vintages, fairness and protected-group impact checks, macroeconomic variables, and a challenger model such as gradient boosting. These extensions require real application and outcome records and should not be evaluated on the synthetic benchmark alone.

Key takeaway. A balanced monitor and an evidence-based escalation process are justified; a claim that one strategy is best for Lending Club is not.

13 References

This section lists the data, software, and methodological references used by the supplied project materials.

1. George, Nathan (curator). *All Lending Club Loan Data*. Kaggle dataset, `https://www.kaggle.com/datasets/wordsforthewise/lending-club`. CC0 Public Domain Dedication. The raw data file was not included with the supplied project package.
2. R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. Project materials indicate R 4.4+; exact runtime version: `[INSERT R VERSION]`.
3. Wickham, H. et al. *Welcome to the tidyverse*. Journal of Open Source Software, 4(43), 1686, 2019.
4. Grolemund, G. and Wickham, H. *Dates and Times Made Easy with lubridate*. Journal of Statistical Software, 40(3), 1–25, 2011.
5. Robin, X. et al. *pROC: an open-source package for R and S+ to analyze and compare ROC curves*. BMC Bioinformatics, 12, 77, 2011.
6. scorecard package for R. Weight-of-evidence binning, scorecard construction, and score application. Package version: `[INSERT SCORECARD VERSION]`.

7. Siddiqi, N. *Credit Risk Scorecards: Developing and Implementing Intelligent Credit Scoring*. 2nd ed., Wiley, 2017. Reference for scorecard development concepts.

A Glossary

This appendix defines the principal technical terms used in the report.

AUC Area Under the ROC Curve; a summary of ranking performance over thresholds.

Bad rate Share of cases with the defined adverse outcome; here the synthetic target or, in the intended raw data, Charged Off status.

Calibration Agreement between predicted probabilities and observed outcome frequencies.

CSI Characteristic Stability Index; distribution shift for an individual predictor.

Discrimination
Ability to rank bad outcomes as riskier than good outcomes.

IV Information Value; a summary measure of how well a binned variable separates good and bad cases.

KS Kolmogorov–Smirnov statistic; maximum distance between cumulative score distributions for goods and bads.

LGD Loss Given Default; assumed loss fraction when a loan defaults.

OOT Out-of-time; chronological evaluation on later observations.

PDO Points to Double the Odds; score increment corresponding to halving default odds.

PD Probability of Default; model-estimated probability of a bad outcome.

PSI Population Stability Index; score-distribution shift relative to a reference group.

Right-censoring
Incomplete outcome observation because the event window has not fully elapsed.

WOE Weight of Evidence; log ratio of good and bad distributions within a bin.

B Reproducibility

The project is implemented in R using tidyverse, lubridate, scorecard, pROC, and supporting packages. The source files are staged as `01_load_clean.R` through `06_remediation_strategies.R`, with `run_pipeline.R` as the end-to-end driver. To reproduce empirical results, place `accepted_2007_to_2018Q4.csv.gz` in the project root or `data/`, then run the pipeline from the project directory. Without that file, the pipeline generates its synthetic fallback.

Listing 2: Reproduction command from the project directory.

```
source("run_pipeline.R")
rmarkdown::render("time_machine.Rmd")
```

The project README indicates R 4.4 or later; the exact R and package versions used to create the supplied artifacts were not recorded in the printed materials. Insert them as `[INSERT R VERSION]` and `[INSERT PACKAGE VERSIONS]` before treating the work as fully reproducible. Seed 42 is set in

the pipeline. Raw borrower-level data are not reproduced in this report; use the Kaggle source link in the references.

Key takeaway. The workflow is rerunnable, but reproducing the actual included benchmark requires the cached RDS objects; reproducing Lending Club results requires the absent raw Kaggle file and recorded environment versions.

C Full results and unresolved placeholders

This appendix gathers the result tables and identifies every value that still needs source output. Tables 1, 3, 5, 6, 7, and 8 are the full printed numeric result tables available in the supplied HTML report. The CSI heat map is Figure 8; a separate CSI export was not printed.

Placeholders requiring real project outputs

- `[INSERT 2015 KS]`: compute/confirm from the 2015 validation output.
- `[INSERT BIN/POINTS TABLE FOR TOP VARIABLES]`: export boundaries, counts, WOE, and points from the fitted scorecard object.
- `[INSERT MODEL COEFFICIENTS AND STANDARD ERRORS]`: export the fitted logistic model summary.
- `[INSERT BIN COUNTS, BOUNDARIES, AND WOE EXAMPLE]`: supply real bin frequencies for the methodology example.
- `[INSERT PROFIT AS % OF AVERAGE LOAN AMOUNT]`: provide the corresponding mean loan amount or calculate the requested normalized profit metric.
- `[INSERT R VERSION]` and `[INSERT PACKAGE VERSIONS]`: record the execution environment.
- `[INSERT FULL CSI TABLE FROM data/csi_tbl.rds]`: include exact per-year, per-feature values if a table is required in addition to Figure 8.

Key takeaway. The source package's benchmark values are transcribed without altering its tables or chart assets; the placeholders above identify outputs absent from the delivered printout or environment metadata.